Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Full-Stack AI Product · Python · FastAPI · React · TypeScript · RAG Pipelines · MCP · Claude Code · Production Ownership · 9yr

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Arlington, Virginia, USA

PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience who builds things end-to-end and owns them in production - not just in pull requests. I use Claude Code as my primary engineering tool daily (code generation, architecture review, test generation, debugging), which Evinova's Study Designer team explicitly names as part of their AI tooling stack. AgentProbe demonstrates the full-stack AI product pattern at production scale: 15-endpoint FastAPI backend, LangGraph orchestration, 5-metric observability engine with structured tracing across all agent steps (LangFuse-equivalent), and a CI-integrated regression gate - built end-to-end using AI-augmented development workflows. LeaseI shows RAG pipelines and MCP in a complex-domain product: a document intelligence and investigation platform where every AI answer is either cited or explicitly flagged as unsupported - deterministic logic makes every regulated call. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime - complex domain logic (affordable housing compliance calculations, regulatory imports, eligibility pipelines) with zero compliance violations, responsible for code in production across 120+ Agile sprints. Clinical trial design is a specialized space I'd bring that same production rigor and genuine curiosity to. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Engineering & Full-Stack Product: Python (9yr), TypeScript, React.js, Claude Code (daily), AI-First Engineering, LangGraph, FastAPI, RAG Pipelines, MCP, Multi-Agent Orchestration, LangChain, ChromaDB, Embeddings, Semantic Search, LLM Integration, Prompt Engineering, LLM Evaluation, LLM-as-Judge, Output Validation, Hallucination Detection, HITL Guardrails, Anthropic API, OpenAI API, Ollama, Observability, Distributed Tracing, Structured Logging

Backend, APIs & Data: FastAPI, Flask, REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Distributed Systems, PostgreSQL, MySQL, Oracle, Redis, SQL, NoSQL, High-Throughput Systems, Performance Optimization, Query Optimization, Indexing, Reliability Engineering, Idempotency, SQLAlchemy, Docker

Cloud, DevOps & Engineering Excellence: AWS (S3, EC2, RDS, Glue, Lambda), Docker, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Performance Testing, Regression Testing, Monitoring, Alerting, Incident Response, Root-Cause Analysis, Runbooks, Design Documents, Technical Mentoring, 300+ Code Reviews, Agile/Scrum, Production Ownership, Linux, Bash, Git

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer - Full-Stack AI Product Delivery

- Designed and shipped a production AI application (LangGraph, FastAPI, Python) as sole technical owner using Claude Code throughout - AI-augmented architecture design, code generation, test generation, and debugging; delivered evaluation pipeline, CI validation gate, HITL guardrails, and observability infrastructure to production in weeks.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977) on AWS; designed observability dashboards improving research comprehension 40%; applied Isolation Forest anomaly detection across 3 research pipelines; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - PropTech / Real Estate SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Production ownership at scale** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ users and 1M+ property records; responsible for code in production through incidents, regulatory changes, and annual release cycles - not just pull requests; RabbitMQ async event-driven processing with retry logic and idempotency guarantees; 200+ PostgreSQL/MySQL/Oracle query optimizations (30% performance gain); Docker microservices redesign (50% scalability improvement); GitHub Actions CI/CD; AWS S3/EC2/RDS/Glue infrastructure.
- **Complex domain logic with zero compliance violations** - owned HUD affordable housing compliance pipelines (eligibility, rent calculations, HAP, tax-credit) with regulatory import accuracy across 7 years of ownership; built and maintained structured logging, monitoring, and alerting infrastructure for production oversight; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%.
- **Technical leadership & code quality** - wrote design documents and architecture decisions for 20+ production modules; mentored 5+ engineers on system design and production best practices; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload; shipped 20+ full-stack modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI Evaluation & Observability Platform - Full-Stack AI Product

FastAPI · LangGraph · RAG · Observability · CI Gate · Claude Code · Open Source

Python · FastAPI · LangGraph · React.js · TypeScript · ChromaDB · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production AI-powered platform built end-to-end using Claude Code - the same AI development tooling Evinova's Study Designer team uses daily. **Full-stack product:** 15-endpoint FastAPI backend, React.js + TypeScript frontend, multi-provider LLM routing (Anthropic, OpenAI, Ollama). **RAG pipelines:** ChromaDB semantic retrieval with embedding-based search - directly in the Evinova AI infra stack. **Observability:** structured tracing across all agent steps with span capture, quality drift monitoring, and alerting - LangFuse-equivalent observability built from first principles. **CI regression gate:** GitHub Actions integration blocks quality degradation before deployment; 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency) across 1,240+ test cases, zero regressions.

Leasel: RAG Pipelines & MCP in Complex-Domain AI Product

RAG Pipelines · MCP · LangGraph · HITL · Production · Domain Complexity

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI platform for a complex regulated domain - directly analogous to clinical trial design: specialized rules, regulated decisions, and domain expertise that takes time to acquire. **RAG pipelines:** section-aware document ingestion, TF-IDF retrieval with hybrid reranking, citation-backed outputs with 100% citation correctness and 0% unsupported-claim rate on eval set. **MCP integration:** live HUD income-limit API client via MCP - the same protocol Evinova lists in its AI/ML infra stack. **HITL guardrails:** deterministic logic makes every regulated call; LLM reasons and extracts, never decides compliance outcomes. **8-agent LangGraph pipeline** (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) - 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Full-Stack AI Application - React.js, TypeScript, Python & AWS

Full Stack · React.js · TypeScript · AWS Lambda · Python · HIPAA · Production

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application in a regulated healthcare environment - React.js + TypeScript frontend with Python REST API backend; LLM-powered NLP translating natural-language queries (EN/ES) into structured FHIR API calls; AWS Lambda serverless backend; PostgreSQL data services; HIPAA-compliant RBAC with complete audit trail and observability stack; Docker containerization. Demonstrates the full-stack capability (React.js + TypeScript UI, Python/FastAPI backend, AWS cloud, production compliance) directly applicable to Evinova's Study Designer product. ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017